PROJECT CHARTER	
Project Name	**Ecosystem Discovery Application**
Date Produced	27 January 2026
Project Goals	The purpose of this project is to educate children in Grades 2–3 (approximately ages 7–8) about Saskatchewan freshwater ecosystems through an interactive, browser-based learning experience. The application introduces freshwater animals commonly found in Saskatchewan, explains basic conservation concepts, and demonstrates how human actions can both negatively and positively impact these ecosystems. The system uses a timed, hint-based discovery game, a visual sticker-book collection system, and support educational videos to encourage engagement and reinforce learning. The project aims to help young learners understand practical ways they can help keep freshwater environments healthy while making the learning process enjoyable, interactive, and age-appropriate.
Project Objectives	• Design a single Saskatchewan freshwater ecosystem as the interactive game environment • Create six freshwater animal representations used in gameplay • Store animal and hint data dynamically using a backend database (Supabase) • Implement a timed, hint-based discovery game where users match animals using clues • Display collected animals in a visual “sticker book” interface • Provide age-appropriate educational hints and facts written at a Grade 2–3 reading level • Include animated interactions (fish movement, card animations, and visual feedback) • Implement a timer system with bonus interactions (e.g., trash collection for extra time) • Provide additional learning through an educational videos page • Ensure a simple, intuitive, and child-friendly user interface using a 2D visual design
Project Budget	The team does not intend to spend any money on the creation of this project. The application is developed using freely available tools and resources, including HTML, CSS, JavaScript, and Supabase for backend data storage. All assets and libraries used are free or open-source.

Project Sponsor	Dr. Tim Maciag, ENSE 281 instructor
Project Manager	Rayansh Chowatia
Additional Key Project Stakeholders	

[The names of key stakeholders that are known at this point in the project, including their job title or project role]	
• The project team: Jeremiah Onunkwo, Rayansh Chowatia, Abrianna Primavera, Aubin Chriss Izere • Dr. Tim Maciag, ENSE 281 instructor • The intended end users: the children who will play our game • Other ENSE 281 student groups who may provide informal peer feedback during project reviews but do not have decision-making authority	
Overall Project Milestones	**Dates**
[A list of the key milestones that are known at this point in the project]	[Milestone dates]

Project Initialization: • Pitch presentation • Business case document • Charter document	Jan 30, 2026
Possible “work-in-progress” meeting	Feb 25, 2026
Project Prerequisites and System Design: • Stakeholder register • Project roles and responsibilities document • Project scope document • GitHub Project Board • Defined and rationalized Minimum Viable Product (MVP) features • Diagrams that demonstrate how the team is using MVC and domain-driven design • Lo-fidelity and/or high -fidelity prototypes • User questionnaire • Project presentation • Self-reflection	March 4, 2026
Project progress scrum/check-in: • Project status report • Project presentation • Self-reflection	March 18, 2026
Project delivery and closing: • Project MVP code • GitHub Project Board • Project installation requirements and instructions • Final project vlog demonstration • Final project report • Final project presentation • Self-refection	April 8, 2026

Overall Project Risks
[A list of the overall risks that are known at this point in the project]

- There is a risk that the project may not be completed within the planned timeline due to academic workload and limited development time.
- The team may need to learn and use tools or technologies that have not been previously covered in class, which could increase the time required for development.
- There is a risk that the intended audience of elementary school children may not find the application sufficiently engaging if the design and interactions are not appealing.
- Some team members have limited experience in visual or artistic design, which may impact the overall visual quality of the application.
- There is a risk that some educational information about freshwater animals may be incomplete or inaccurate if not carefully researched and validated.
- There is a risk that the project scope may expand beyond a manageable level if additional ecosystems, animals, or complex animations are introduced.
- There is a risk that the interaction flow (right/wrong answer behavior and sticker unlock logic) may not be clearly defined, which could impact usability and user understanding.